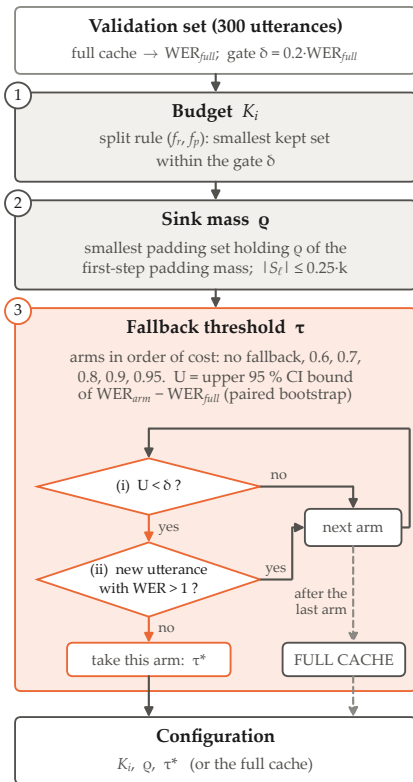


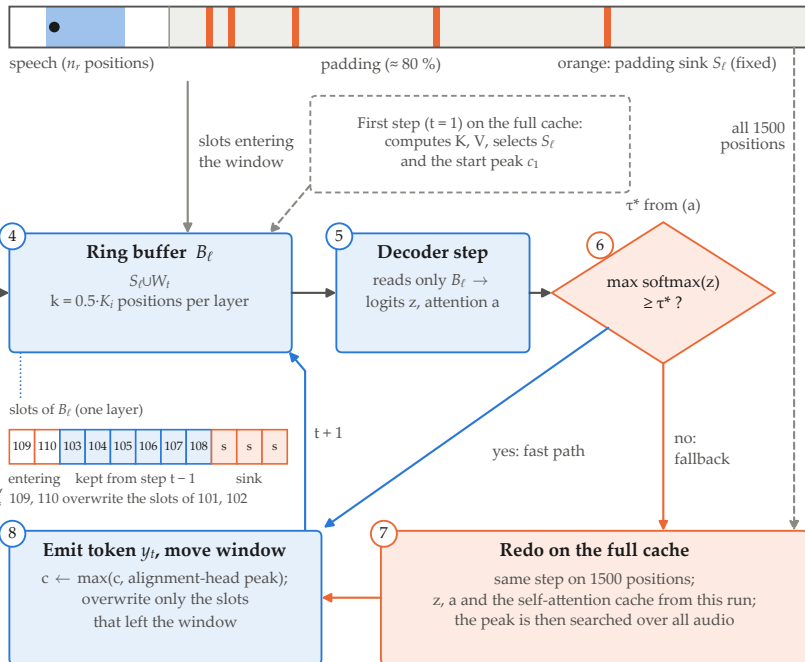
(a) Calibration, once per model



(b) Decoding step t

Full cross-attention cache K, V — 1500 positions per layer (stays in DRAM)

blue: window $W_t = [c_t - 0.1 \cdot w_\ell, c_t + 0.9 \cdot w_\ell]$, moves with decoding; $\bullet$ = alignment peak c_t



PadSink-Track fast path

padding sink / fallback

$\rightarrow$ every step / every arm

$\dashrightarrow$ conditional: first step, fallback, last arm